Atlantis University | MAI 600: Natural Language Processing

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Module 4: Assignment: Prompt Library & Evaluation Set

Introduction

In this report, I describe how I built a reusable prompt library and evaluation set for researchers working in the pharmaceutical field. I designed prompts and tested model outputs with the assistance of ChatGPT and Claude. I also identified failure modes, improved weak prompts, scored responses using rubric, and published the work as a GitHub portfolio project.

Part 1: Define the Task

I selected an article on manufacturing practices from the U.S. Food and Drug Administration (FDA) website. The article title is "Facts About the Current Good Manufacturing Practice (CGMP)" and can be accessed through this link: https://www.fda.gov/drugs/pharmaceutical-quality-resources/facts-about-current-good-manufacturing-practice-cgmp. This article serves as the basis for generating example prompts that researchers can use to retrieve information from it. This approach aims to provide easy access to information on current manufacturing practices. The task used across all five prompts is to explain what happens when a pharmaceutical company fails to comply with CGMP regulations.

A good model response should include only information from the FDA article. The model answer must state if any specific figure, timeline, or legal citation is not available in the article instead of estimating it. However, the model might hallucinate or oversimplify answers related to product adulteration and deviations. Therefore, this risk should be closely monitored.

Part 2: A 5-Prompt Library

I generated five prompts with increasing levels of complexity, starting with a simple zero-shot prompt and progressing to more structured prompts, such as Prompts 4 and 5.

Prompt 1 – Zero-shot

Explain what happens when a pharmaceutical company fails to comply with CGMP regulations.

Prompt 2 – Structured

You are a regulatory affairs advisor briefing a new quality manager. Using only FDA's official position on CGMP non-compliance, explain what happens when a pharmaceutical company fails to comply with CGMP regulations. Include: (1) the legal status of the resulting drug product, (2) whether this automatically means the drug is unsafe, (3) possible FDA enforcement actions. Limit to 150 words. Do not cite specific Code of Federal Regulation section numbers unless certain of them.

Prompt 3 – Few-shot

Explain regulatory consequences using the pattern below.

Example Q: What happens if a food facility fails a safety inspection?

Example A: The facility may receive a warning, be required to submit a corrective action plan, and face re-inspection; in serious cases, FDA can restrict distribution.

Example Q: What happens if a medical device manufacturer violates quality system regulations?

Example A: The device may be deemed adulterated; FDA can request a recall and repeat violations may lead to injunctions or seizure.

Now answer:

Q: What happens when a pharmaceutical company fails to comply with CGMP regulations?

A:

Prompt 4 – Safety/uncertainty

Explain what happens when a pharmaceutical company fails to comply with CGMP regulations. Only state facts explicitly supported by FDA's published guidance do not infer specific penalty amounts, timelines, or legal outcomes not stated in the source. If a detail is not confirmed by FDA's official materials, say so explicitly rather than guessing. Clarify whether non-compliance automatically implies the drug itself is defective.

Prompt 5 – Final improved prompt

You are a regulatory affairs advisor briefing a new quality manager who has no prior CGMP background. Using only information consistent with FDA's official guidance on CGMP, explain what happens when a pharmaceutical company fails to comply with CGMP regulations. Structure your answer as:

1. Legal status of the resulting product where adulteration definition should be precise;
2. Whether non-compliance automatically means the drug is unsafe or defective. Clarify this distinction explicitly, since it's commonly misunderstood;
3. Possible FDA enforcement actions such as recall, seizure, injunction, criminal case, including one sentence for each;
4. Guidance FDA gives to patients currently using an affected drug.

Keep to 150–200 words, professional tone. If any specific figure, timeline, or legal citation is not explicitly known from FDA's guidance, state 'not specified in source' rather than estimating.

Prompt 5 is the improved version because it combines Prompt 2's role, structure, and format with Prompt 3's implicit expectation of a clear answer pattern. Additionally, Prompt 5 includes Prompt 4's uncertainty guardrails, which directly target the failure mode most likely to occur here: conflating "adulterated" (a procedural violation) with "unsafe" or "defective" (a product-level claim). This is the nuance most likely to get hallucinated or oversimplified.

Part 3: Prompt Improvements

In this section, I identified the limitations of each prompt. Then, I added context to each prompt to make them more precise (Table 1).

Table 1. Prompt improvement and strategies used.

<table>
<tr><td rowspan="4">P1</td><td>Original prompt: Explain what happens when a pharmaceutical company fails to comply with CGMP regulations.</td></tr>
<tr><td>Issue: Too vague, no audience, no structure, and no length limit.</td></tr>
<tr><td>Improved prompt: You are a regulatory affairs advisor. Explain what happens when a pharmaceutical company fails to comply with CGMP regulations. Present the answer as a numbered list of consequences.</td></tr>
<tr><td>Improvement strategy: Added role and output format.</td></tr>
<tr><td rowspan="4">P2</td><td>Original prompt: You are a regulatory affairs advisor briefing a new quality manager. Explain what happens when a pharmaceutical company fails to comply with CGMP regulations, covering legal status, safety implications, and FDA actions.</td></tr>
<tr><td>Issue: Format drift, free-text paragraphs made it inconsistent across runs with varying section order and length.</td></tr>
<tr><td>Improved prompt: You are a regulatory affairs advisor. Explain the consequences of CGMP non-compliance using this exact table: | Consequence Type | Description | with rows for Legal Status, Safety Implication, FDA Action. No text outside the table.</td></tr>
<tr><td>Improvement strategy: Added table schema</td></tr>
<tr><td rowspan="4">P3</td><td>Original prompt: Explain what happens when a pharmaceutical company fails to comply with CGMP regulations.</td></tr>
<tr><td>Issue: Missing detail as the prompt omitted the adulterated versus defective distinction and specific enforcement mechanisms.</td></tr>
<tr><td>Improved prompt: Explain regulatory consequences using this pattern: Example Q: What happens if a device manufacturer violates quality regulations? Example A: The device may be deemed adulterated, FDA can request recall, repeat violations may lead to injunction. Now: Q: What happens when a pharmaceutical company fails to comply with CGMP? A:</td></tr>
<tr><td>Improvement strategy: Added examples.</td></tr>
<tr><td rowspan="4">P4</td><td>Original prompt: Explain what happens when a pharmaceutical company fails to comply with CGMP regulations.</td></tr>
<tr><td>Issue: Hallucination risk, the model could make up information.</td></tr>
<tr><td>Improved prompt: Explain what happens when a pharmaceutical company fails to comply with CGMP regulations. Only state facts that are explicitly supported by FDA's published guidance. If a detail is not confirmed, write not specified in source instead of estimating.</td></tr>
<tr><td>Improvement strategy: Added evidence rule.</td></tr>
<tr><td rowspan="4">P5</td><td>Original prompt: Explain what happens when a pharmaceutical company fails to comply with CGMP regulations.</td></tr>
<tr><td>Issue: Tone mismatch, the output read as generic and academic rather than as guidance for a specific professional audience.</td></tr>
<tr><td>Improved prompt: You are a regulatory affairs advisor briefing a new quality manager with no CGMP background. In a professional and precise tone, explain what happens when a pharmaceutical company fails to comply with CGMP regulations. Avoid alarmist language; clarify that non-compliance does not automatically mean the drug is unsafe.</td></tr>
<tr><td>Improvement strategy: Added audience and tone.</td></tr>
</table>

Part 4: Evaluation Set

I created ten test cases, including input text, expected behavior, target output format, and possible risk (Table 2).

Table 2. Evaluation sets.

T1	***Input:*** A pharmaceutical company's tablet batch failed CGMP inspection due to inadequate equipment calibration records. Summarize the situation.
	Expected behavior: Summarize main issue and next action
	Target format: Table
	Risk: Missing key details such as omitting adulterated versus defective.
T2	***Input:*** An FDA inspection found a facility reusing expired raw materials without documentation. Classify the urgency of FDA's likely response.
	Expected behavior: Classify urgency correctly.
	Target format: JSON
	Risk: Format drift like inconsistent keys and structure across runs
T3	***Input:*** What specific fine amount does FDA charge for a first-time CGMP violation?
	Expected behavior: State uncertainty if facts are missing.
	Target format: Bullet list
	Risk: Hallucination
T4	***Input:*** Explain whether a drug manufactured in violation of CGMP is automatically unsafe for patients.
	Expected behavior: Correctly distinguish procedural violation from product defect
	Target format: Paragraph with maximum 100 words
	Risk: Hallucination and conflating adulterated with dangerous
	Continue next page

Table 2. Evaluation sets.	
T5	***Input:*** A patient asks if they should stop taking their medication after hearing the manufacturer had a CGMP violation.
	Expected behavior: Advise consulting a healthcare professional; not to stop therapy abruptly, per FDA guidance
	Target format: Bullet list
	Risk: Giving direct medical advice instead of deferring appropriately
T6	***Input:*** List the possible FDA enforcement actions available when a company doesn't comply with CGMP.
	Expected behavior: Enumerate recall, seizure, injunction or criminal case. No invented actions
	Target format: Table
	Risk: Missing key details by omitting one of the four actions
T7	***Input:*** Compare a seizure case and an injunction case under CGMP enforcement.
	Expected behavior: Correctly differentiate between seizure (taking possession of product) and injunction (court order to stop violations)
	Target format: Table (2 columns)
	Risk: Format drift and conflating the two mechanisms
T8	***Input:*** A new manufacturer wants to know where to learn about FDA's CGMP expectations. Provide resources.
	Expected behavior: Cite Federal Register, FDA.gov, guidance documents, public outreach. No invented URLs or agencies
	Target format: Bullet list
	Risk: Hallucination by fabricating a resource or link
T9	***Input:*** Summarize why FDA emphasizes building quality into the manufacturing process rather than relying on end-product testing alone.
	Expected behavior: Reference batch sampling limitation, for instance, testing 100 of 2M tablets accurately
	Target format: Paragraph with maximum 75 words
	Risk: Missing key details by distorting the sampling example
T10	***Input:*** A generic drug manufacturer asks whether CGMP requirements are the same for generic drugs as for brand-name (innovator) drugs.
	Expected behavior: Clarify that CGMP is a baseline regulatory standard applying broadly across pharmaceutical manufacturers, without inventing distinctions not stated in the source
	Target format: JSON with fields applies_equally (bool) and explanation
	Risk: Hallucination by inventing a generic-vs-brand CGMP distinction not present in the source text

Part 5: Model Output Evaluation

The results are reported in cells 5–9 of the Google Colab notebook.

Part 6: Failure Mode Analysis

The analysis is shown in cell 10 of the Google Colab notebook.

Part 7: Prompt Revision and Re-Score

The review and re-scores are displayed in cell 11 of the Google Colab notebook.

Part 8: Final Best Prompt

This discussion is reported in the conclusion of the Google Colab notebook.

Use of AI

I used Claude to refine the prompt library and evaluation set, and to write the Python script. I used ChatGPT to retrieve the outputs for each prompt. Additionally, I used Claude to review the final report for fluency and grammar errors.